

AI eksamensopgave - Car Dealership

GitHub Link

<https://github.com/Peter537/Soft1-AI-Eksamensopgave>

Group F

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Business case / Problem statement

Across the world, cars are one of the most essential modes of transportation. In some regions more than others, which leads to a high volume of vehicle transactions. For example, according to S&P Global:

"[...] 2025 US auto sales in April expected to reach 1.49 million units [...]"

Source: [automotive-insights](#) (last visited: 29-04-2025).

Where there is a high number of cars being bought and sold, they will eventually end up at used car dealerships. According to [IBISWorld](#) (last visited: 29-04-2025): There were

"There was 149,002 Used Car Dealers in the US businesses as of 2024, an decrease of -1.7% from 2023."

Based on this, we can wonder:

- How can these used car dealerships price their cars?
- What factors impact the pricing of a used car?
- Can machine learning be used to provide an accurate price prediction?
- Can a Chatbot accurately answer user questions for the car dealerships?

Hypothesis

We believe that we can create a machine learning model that can predict the price of a used car based on its attributes. We also believe that we can create a chatbot that can answer user questions about the car dealership and its cars based on the dataset from part1.

Part 1

Our Jupyter notebook is located in [/part1/part1.ipynb](#).

We have documentation inside the Notebook.

Part 2

Our setup is documented in the README.md file located in [/part2/README.md](#).

The solution can be run by executing `streamlit run app.py` in the terminal inside the [/part2](#) folder.